

Beyond Accuracy: Explainable Trustworthy Index and Explainable Robustness Index for Evaluating CNN

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ARTICLE INFO

Keywords:

Explainable AI (XAI), Trustworthiness, Robustness, CNN, Grad-CAM, Perturbation Analysis, Plant Leaf Classification

ABSTRACT

Deep neural networks often achieve high classification accuracy. Despite high accuracy, the reliability of CNN models remains a concern. Sometimes CNN models may rely on background or irrelevant features, undermining trustworthiness. Besides, with the use of perturbations, models' accuracy and object reliance may decline. Such degradation raises questions about their robustness. In this research, we have proposed two novel evaluation metrics: the **Explainable Trustworthy Index (ETI)** and the **Explainable Robustness Index (ERI)**. ETI combines accuracy with the **Localization Mask Overlap Score (LMOS)**, which measures the proportion of Grad-CAM salient pixels that fall within the ground-truth object regions. A model with high ETI clearly indicates that both its accuracy and object reliance are high and satisfactory. ERI extends this to perturbed inputs by jointly evaluating accuracy stability and LMOS stability. Like ETI, a model with high ERI indicates that both its accuracy and object focus remain stable under the use of perturbations. We have used eight CNN architectures on four datasets and nine perturbations (e.g., noise, blur, motion blur, occlusion) to calculate ETI and ERI. We have demonstrated that high accuracy sometimes does not guarantee high ETI or ERI. For instance, on the EPLI dataset, Custom CNN has achieved 99.45% accuracy but the lowest ETI (0.5922) due to poor object localization. It has also provided the lowest ERI. ETI and ERI provide a more holistic assessment of model reliability for safety-critical applications.

1. Introduction

Convolutional Neural Networks (CNNs) have achieved remarkable performance in a wide range of domains such as medical imaging, agriculture, and astronomy, where predictions support important decision-making processes [1, 2, 3, 4]. In such applications, classification accuracy is an essential performance measure, but accuracy alone does not necessarily indicate whether a model is reliable. A CNN model is considered more reliable when its predictions are grounded in task-relevant visual features rather than spurious correlations [5]. Therefore, beyond predictive accuracy, models must be quantitatively evaluated to determine whether their predictions depend on task-relevant regions rather than irrelevant background cues. Despite excellent accuracy, a model may not provide reliable predictions. Explainability methods, such as Grad-CAM [6], show that a model may rely mainly on background and irrelevant regions instead of focused objects [7, 8]. This raises questions about reliability and trustworthiness.

Model behavior can shift due to real-world perturbations e.g., *noise*, *blur*, *lighting shifts*, *compression*, and *occlusion*. These distortions can alter both CNN accuracy and the visual regions driving predictions. Consequently, even

when a model maintains relatively high accuracy under perturbations, its underlying visual explanations may still become inconsistent [5, 9]. Conventional robustness evaluation methods are primarily based on accuracy change under perturbation. They do not capture whether the model maintains consistent reliance on task-relevant regions under perturbed conditions. These observations motivate an evaluation framework that considers both predictive behavior and explanation behavior.

Explainable Artificial Intelligence (XAI) provides a framework for examining visual evidences used by CNNs for their predictions. In particular, Grad-CAM generates class-discriminative heatmaps that highlight image regions contributing to a model's predictions [6, 10]. We refer to the Grad-CAM heatmaps as explanations of the corresponding model's predictions. Such explanations can reveal whether a model achieves high classification accuracy by focusing substantially on background or other irrelevant regions rather than target objects. Previous studies have shown that background information and spurious correlations can influence deep neural network decisions and may lead to shortcut learning [11, 12, 13]. Consequently, a highly accurate model does not confirm that the predictions are made on appropriate object-level information. This creates an important distinction between predictive performance and explainable reliability. Motivated by these observations, we propose metrics that integrate accuracy with object focus.

The gap of existing studies

Existing works have developed diverse approaches for evaluating CNN reliability, robustness, and explanation

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quality described in Section 7. Reliability has been examined through uncertainty estimation, calibration, confidence, failure detection, risk-coverage analysis, and broader trustworthiness criteria. Robustness has been evaluated through performance degradation under common corruptions, natural distribution shifts, and other perturbations, while some studies have also investigated the stability of explanations under corrupted inputs. Similarly, explanation quality and object reliance have been studied using faithfulness-based evaluation, ground-truth explanations, and spatial correspondence between explanations and relevant image regions. These studies demonstrate that reliability, robustness, and explanation quality are multidimensional concepts and can be quantified from different perspectives. However, the existing evaluation approaches generally emphasize one or a subset of these dimensions, and they do not provide a unified model-level evaluation that simultaneously considers predictive accuracy, the spatial focus of decision-relevant visual evidence, and the stability of both under input perturbations. In particular, a model may exhibit high predictive accuracy or relatively stable accuracy under perturbations while relying on visual evidence that is poorly aligned with the target object, or its visual focus may change substantially even when its predictive performance remains relatively stable. This motivates a complementary evaluation framework that jointly examines predictive performance, object-focused explanation quality, and explanation-aware robustness.

We propose two novel evaluation metrics: Explainable Trustworthy Index (ETI) and Explainable Robustness Index (ERI) using XAI methods. Our objective is not to modify or improve CNN models, rather to provide quantitative measures for evaluating and comparing their reliability and robustness. ETI measures baseline reliability on unperturbed images by jointly evaluating classification accuracy and object-focused explanation quality. In contrast, ERI measures robustness to input perturbations by assessing whether accuracy and explanation quality remain consistent with clean-image behavior. In essence, ETI and ERI address two different questions: ETI asks “Whether a model is accurate and appropriately focused on the target object under normal conditions?”, whereas ERI asks “Whether these properties remain stable when the input is perturbed?”

Figure 1 illustrates the distinct advantages of our XAI-based metrics over traditional accuracy. Here, we show the predictions of two models - Xception and a custom CNN (detailed in Section ??) - on the same sample test image. Both models were trained on identical training data for a leaf classification task, with each achieving an accuracy of 99.45% on the validation set. However, evaluating the models using our proposed metrics reveals that Xception achieves significantly higher ETI and ERI scores (0.88 and 0.93) compared to the custom CNN (0.59 and 0.79). Inspecting the Grad-CAM heatmaps confirms that the custom CNN relies heavily on background rather than the leaf object itself. While standard accuracy fails to identify this flaw, our proposed metrics correctly indicate that selecting the custom CNN would lead to sub-optimal model deployment.

We introduce the Localization Mask Overlap Score (LMOS) to quantify object-focused explanation quality which is the proportion of the model’s decision-relevant activation that overlaps with the annotated target-object region. LMOS is used in both ETI and ERI calculation. It is observed that in Grad-CAM visualizations, red and yellow regions are activated most i.e., red and yellow have greater influence than other colors on models’ predictions [14, 15]. We integrate this principle into LMOS; a higher LMOS value indicates that a larger proportion of the model’s highlighted regions falls within the target object boundary rather than the background.

We comprehensively evaluate the proposed metrics using four real-world datasets comprising two leaf image datasets: EPLI [16], BDML[17] and two flower image datasets: FLRS [18], FR [19]. Eight CNN architectures are trained and evaluated on the datasets. We demonstrate that a custom CNN model achieves 99.45% classification accuracy on the EPLI dataset, yet its mean LMOS is only 0.1900, resulting in the lowest ETI score of 0.5922 among the evaluated models. Its LMOS is only 0.1900, indicating that only 19.00% of its decision-relevant activation, on average, falls within the annotated target-object regions, while the remaining 81.00% lies outside these regions. In contrast, the MobileNetV2 model achieves 99.45% accuracy (same as the custom CNN model) but obtains a substantially higher mean LMOS of 0.7765, indicating that 77.65% of its decision-relevant activation falls within the target-object regions, with only 22.35% outside them and an ETI of 0.8855. This demonstrates that models with similar high accuracy can exhibit substantially different levels of object-focused reliance and the model with higher LMOS should be selected for real-world deployment. Importantly, a low LMOS score does not inherently denote shortcut learning or overfitting, but rather quantifies the extent to which a model’s salient activations diverge from the target object.

The proposed ERI combines *accuracy stability* and *LMOS stability* to provide a joint measure of explainable robustness. This distinction is important because predictive stability and explanation stability are not correlated. For example, in our experiments we observe that the Custom CNN model under contrast perturbation maintains relatively high accuracy stability of 0.8494 but has substantially lower LMOS stability of 0.4545 on the EPLI dataset. This indicates that, although its predictive performance remains comparatively stable, its object-focused explanation changes substantially relative to the clean-image condition. Conversely, the ResNet50 model exhibits relatively low accuracy stability of 0.4599 but higher LMOS stability of 0.7830 under the same perturbation, indicating that its object-focused explanation remains comparatively more consistent despite a larger change in predictive performance. These demonstrate that accuracy stability alone cannot fully characterize robustness: a model may preserve its predictive performance while substantially changing its object-focused behavior, or it may preserve its visual focus while its predictive performance deteriorates.

The significance of ETI and ERI lies in their ability to capture key dimensions of CNN reliability and robustness that standard accuracy metrics overlook. However, a high ETI or ERI score should not be equated with overall model trustworthiness, which encompasses broader factors beyond the scope of this work. Rather, these metrics focus specifically on predictive accuracy, object-focused explanation quality, and their stability under input perturbations - offering complementary criteria for evaluating CNN models beyond conventional accuracy benchmarks.

Two questions that the proposed measures can answer - How trustworthy a model is? Trustworthy means - predictions are based on real objects in the images. How robust a model is? Robustness means - performance of the model will not degrade under perturbations.

To show in exp:

- Acc high but not reliable - may be low robustness or predictions based on BG.
- a model having high acc but low LMOS and has one or more of the mentioned problems
- A model is trustworthy if it has high accuracy and high LMOS - show this. Idea: models with high ACC an high LMOS making more predictions based on objects in images rather than BGs
- Show LMOS still works with perturbations and ACC and other traditional measure fails.
- Justify distinctions between EI and RI
- Show - traditional measures fail under perturbations but proposed measures hold.

2. Problem Definition

Let $D = \{(x_i, y_i, M_i)\}_{i=1}^N$ be a dataset containing N test images, where $x_i \in \mathbb{R}^{n \times m}$ is the i -th image, y_i is its ground-truth label, and M_i is the corresponding ground-truth mask of the target object. Conventional evaluation methods primarily measure the classification accuracy of a model. Let $acc \in [0, 1]$ denote the classification accuracy of a model.

Two CNNs may achieve identical accuracy while relying on substantially different image regions. In particular, a model may achieve high accuracy while its Grad-CAM activation map reveals that the activated regions concentrated outside the target object. Hence, accuracy is insufficient to distinguish models according to their object-focused explanatory behavior.

We seek to quantify the alignment between a model's rationale and its predictions. Let H_i denote the Grad-CAM heatmap corresponding to image x_i and prediction $\hat{y}_i$. We define a scoring function $LMOS_i \in [0, 1]$ to measure the overlap ratio between the decision-relevant regions of H_i and the ground-truth object mask M_i . To indicate model

Table 1

Notation used in problem formulation.

Notation	Description
D	Dataset of N images with labels and target masks
N	Number of images in the dataset
x_i	i -th input image
y_i	Ground-truth class label of x_i
$\hat{y}_i$	Predicted class label for x_i
M_i	Ground-truth binary mask of the target object in x_i
H_i	Grad-CAM heatmap for x_i and $\hat{y}_i$
$LMOS_i$	Object-focused localization score for x_i
$LMOS$	Mean LMOS over the evaluated dataset
acc	Classification accuracy of a CNN
$\mathcal{P}$	Set of P input perturbations
P	Number of evaluated perturbations
S_{acc}	Overall accuracy stability under perturbations
S_{LMOS}	Overall LMOS stability under perturbations
T	Generic evaluation measure based on acc and $LMOS$
R	Generic robustness measure based on S_{acc} and S_{LMOS}
$\Psi(\cdot)$	Function combining accuracy and LMOS
$\Phi(\cdot)$	Function combining accuracy and LMOS stability

trustworthiness and robustness, we address the following two sub problems in this paper.

Firstly, we want to evaluate CNN models considering both their accuracy (acc) and mean object-focused explanation quality (LMOS). Formally, the objective is to develop an evaluation measure T as a function of acc and LMOS: $T = \Psi(acc, LMOS)$. We address this problem by the proposed Explainable Trustworthy Index (ETI).

Secondly, we want to incorporate CNNs behavioural changes in response to perturbations in model evaluation metric. Let $\mathcal{P} = \{p_1, p_2, \dots, p_P\}$ denote a set of input perturbations. A robust CNN should not only preserve its predictive performance under any changes in $\mathcal{P}$ but also maintain a similar object-focused activation pattern similar to the model's behavior on the corresponding clean image. Let S_{acc} and S_{LMOS} represent the overall accuracy stability (the model's stability to perform under perturbation) and LMOS stability (the model's stability to remain the object-focused for making predictions) of a model. Our second problem is to obtain an evaluation measure R that jointly captures these two forms of stability across the evaluated perturbations: $R = \Phi(S_{acc}, S_{LMOS})$. We propose Explainable Robustness Index (ERI) to address this problem. Table 1 describes the notations used in the article.

3. System Overview

4. Explainable Trustworthy Index and Explainable Robustness Index

Both of the proposed TI and RI metrics use LMOS score. LMOS measure the consistency between model-generated explanations and ground-truth object regions. [One liner on TI and RI to be added here](#)

4.1. Localization Mask Overlap Score (LMOS).

need to say LMOS is an instance specific (or local) measure. Other metrics e.g., acc - is a global measure - indicate overall perf. Thats why we use LMOS in TI definition, just need to say it outloud here LMOS calculates how much a model focuses on true object regions (instead of backgrounds) for prediction. Let $H \in \mathbb{R}^{h \times w}$ denote the Grad-CAM heatmap generated from the last convolutional layer of a network. H is resized to the model's input image size. if H is size how can it be $\mathbb{R}^{h \times w}$? The resized heatmap size is 299×299 for Xception, 224×224 for other models. A binary attention map is obtained from H by thresholding:

$$\hat{H}(i, j) = \mathbb{I}(\text{Resize}(H)(i, j) > \tau)$$

where $\mathbb{I}(\cdot)$ is an indicator function and τ is the threshold to or for??. We set $\tau = 0.7$ why? through experiments or what? how $\tau = 0.7$ restricts to yellow and red pixels?? to restrict our LMOS calculations to red and yellow pixels, as the red and yellow pixels exhibit the highest activation levels in model predictions. Hence, $\hat{H}(i, j)$ is a binary mask representing the activated decision-relevant regions.

LMOS is defined as the ratio of activated decision-relevant regions inside the object region and the total activated decision-relevant regions. Mathematically:

$$\text{LMOS} = \frac{\sum_{i=1}^h \sum_{j=1}^w \hat{h}(i, j) M(i, j)}{\sum_{i=1}^h \sum_{j=1}^w \hat{H}(i, j)} \quad (1)$$

where $M \in \{0, 1\}^{h \times w}$ denote the resized ground-truth binary segmentation mask. waht is L here? what is $\hat{h}$ here? how do we get M ???

4.1.1. Trustworthy Index (TI)

In this article, we propose Trustworthy Index (TI) to measure trustworthiness of a CNN model's predictions. We combine the accuracy, explainable consistency and the mean LMOS (Eq. 1) to define TI.

Let acc denote the classification accuracy of a model on test set and $\overline{\text{LMOS}}$ denote the mean LMOS scores across the same test samples. We define TI defined as follows:

$$\text{TI} = \frac{acc + \overline{\text{LMOS}}}{2}.$$

In this study, both acc and LMOS are equally important and are normalized to the range $[0, 1]$. acc measures predictive performance, whereas LMOS measures the mean explanation consistency with the target object. We provide equal weighting to both components to avoid favoring either predictive performance or explanatory quality. need to rewrite

Variations of TI. To demonstrate the impact of LMOS and acc on the proposed TI, we have calculated two variants of TI as folllows:

$$\text{TI}_{\text{w/o LMOS}} = \frac{acc}{2}, \quad \text{TI}_{\text{w/o acc}} = \frac{\text{LMOS}}{2}.$$

4.1.2. Robustness Index (RI)

We estimate the robustness of a model by measure its' performance deviation under perturbation in the test datasets. Let acc_c denote the accuracy of a model on a clean test dataset (not perturbed) and acc_p denote the accuracy obtained on the test set after applying a specific perturbation p (see Sec. 5.5 for all the perturbation methods used in this research). We introduce the Accuracy Stability (S_{acc}) score if not introduced by you cite under a specific perturbation p as:

$$S_{acc}^p = 1 - \frac{|acc_c - acc_p|}{acc_c + acc_p + \epsilon}$$

where ϵ is a small positive constant introduced to avoid division by zero error. A higher value of S_{acc}^p indicates that the model's performance is more consistent under the corresponding perturbed condition p .

To estimate an overall measure of S_{acc} , we compute S_{acc}^p across all $p \in P$ where P is the set of all perturbation methods used in the experiments, see Section 5.5 and take the mean of all S_{acc}^p . This can be defined as:

$$\overline{S}_{acc} = \frac{1}{|P|} \sum_{p=1}^{|P|} S_{acc}^p$$

Similarly, we introduce LMOS Stability to measures how closely a model's object reliance under perturbation remains to its object reliance on clean images. To evaluate the robustness under input perturbations, we have measured the stability of LMOS when input images are perturbed. need rewriting

For each test sample, we have computed LMOS for both the original image and its perturbed version. Let N be the number of test samples in a test set. Let $\text{LMOS}_{\text{clean}}$ denote the LMOS value computed from an original image and $\text{LMOS}_{\text{perturbed}}$ denote the LMOS value computed from the corresponding perturbed image.

To quantify the consistency between these two values, we define the LMOS stability under a perturbation:

$$S_{\text{LMOS}} = 1 - \frac{|\text{LMOS}_{\text{clean}} - \text{LMOS}_{\text{perturbed}}|}{\text{LMOS}_{\text{clean}} + \text{LMOS}_{\text{perturbed}} + \epsilon}$$

, where ϵ is a small constant to avoid division by zero error. why not the denominator is using abs values?

A higher value of S_{LMOS} indicates that the model's reliance on the focus object remains more consistent under the corresponding perturbation. The LMOS stability under a perturbation p is calculated as an average across all N test samples as:

$$S_{\text{LMOS}}^{(p)} = \frac{1}{N} \sum_{i=1}^N S_{\text{LMOS},i}^{(p)} \quad (2)$$

where $S_{\text{LMOS},i}^{(p)}$ denotes the LMOS stability of the i -th test sample under the perturbation p , and N represents the total number of test samples.

To obtain an overall measure of LMOS stability across all evaluated perturbations, the perturbation-specific LMOS stability scores are subsequently averaged as:

$$\bar{S}_{\text{LMOS}} = \frac{1}{P} \sum_{p=1}^P S_{\text{LMOS}}^{(p)} \quad (3)$$

where P denotes the total number of evaluated perturbations. In this study, $P = 9$.

To jointly evaluate the robustness of both classification performance and explainability under input perturbations, we have proposed the *Explainable Robustness Index* (ERI). Unlike accuracy stability and LMOS stability, which separately assess the consistency of predictive performance and model localization behavior, ERI provides a unified measure of robustness by combining the accuracy stability and LMOS stability.

Let s_{acc}^p be accuracy stability and s_{LMOS}^p be LMOS stability under perturbation p . Now, ERI under perturbation p is calculated as:

$$\text{ERI}^p = \frac{s_{\text{acc}}^p + s_{\text{LMOS}}^p}{2} \quad (4)$$

Let $\bar{S}_{\text{acc}}$ denote the overall accuracy stability and $\bar{S}_{\text{LMOS}}$ denote the overall LMOS stability, both obtained by averaging their corresponding perturbation-specific stability scores across the P evaluated perturbations. The proposed ERI is defined as:

$$\text{ERI} = \frac{\bar{S}_{\text{acc}} + \bar{S}_{\text{LMOS}}}{2} \quad (5)$$

Here, $\bar{S}_{\text{acc}}$ represents the overall consistency of classification accuracy between clean and perturbed inputs, while $\bar{S}_{\text{LMOS}}$ represents the overall consistency of the model's reliance on the focus object under perturbations. Therefore, ERI jointly captures two complementary dimensions of robustness: *predictive stability* and *explanatory stability*.

Since both component scores are bounded in the same range, the resulting ERI also provides a normalized measure of overall robustness. A higher ERI indicates that a model simultaneously maintains its classification performance and preserves its object-focused behavior across the evaluated perturbations. Conversely, a lower ERI indicates that the model exhibits greater instability in one or both dimensions. **Variations of RI.** To analyze the contribution of the individual components of RI (i.e., S_{acc} and S_{LMOS}), we calculate two variants by removing one component at a time:

$$\text{RI}_{\text{w/o LMOS}} = \frac{\bar{S}_{\text{acc}}}{2}, \quad \text{RI}_{\text{w/o acc}} = \frac{\bar{S}_{\text{LMOS}}}{2}.$$

5. Experimental Setup

5.1. Dataset

We have used four publicly available datasets in our empirical study of the proposed evaluation measures:

- **EPLI:** EPLI (introduced as EgyPLI in [16]) contains 3,588 JPG images of size 256×256 pixels. There are images of 8 species including apple, berry, fig, guava, orange, plum, persimmon, and tomato. The dataset is annotated for image-level classification [16].
- **BDML:** BDML (also known as BDMediLeaves [17]) is a dataset of 2,029 original leaf images (augmented to 38,606) from 10 medicinal plant species. They are captured with an iPhone 13 under natural light at 3024×4032 resolution. This dataset is designed for machine learning-based species identification. We used the original dataset (not the augmented one) in our experiments [17].
- **FLRS:** FLRS (or Flowers [18]) is a large image dataset for flower classification containing 3,670 training images of five flower species including daisies, tulips, sunflowers, dandelions, and roses. It is available through the TensorFlow Datasets (TFDS) collection [18]. *what is the resolution of the images in this dataset?*
- **FR:** This kaggle [19] dataset is very similar to the FLRS dataset above consisting the same five species (daisies, dandelions, roses, sunflowers, and tulips). However, the kaggle version contains 4,242 JPG images of resolution of 320×240 pixels. Although the image classes are the same as FLRS, the image samples are different. This repository was also used in previous research [20, 21]. *is this different than the above FLRS?*

Specific difference between FLRS and KFLR.

Dataset preparation. The BDML dataset is already splitted into training, validation and testing sets. We split the other three datasets into training, validation and test subsets with an approximate ratio of 80%, 10% and 10%. Table 2 presents the class-wise training, validation, and test splits. *did you preserve the class distributions in the three splits?*

did you do this? or already in the sources? To simplify explainability and localization analysis, the test images have been manually annotated to generate corresponding JSON files. These JSON annotations describe the polygonal boundaries of the target leaf in each image. The JSON files have been converted into binary masks. Here, each mask represents only the target leaf region and excludes the background and other objects in the scene.

5.2. Baseline models and evaluation metrics.

TBD

5.3. Classification Performance

5.4. Grad-CAM Visualization

Grad-CAM visualization shows that CNN models often predict correctly but they sometimes rely on the background or non-object regions. Besides, under perturbations, their reliance on object pixels hugely changes. *it is not necessary here. We may include the implementation of grad-cam in the method implementation section above*

Table 2

Class-wise distribution of the experimental datasets.

Dataset / Class	Total	Train	Val	Test
<i>EgyPLI</i>				
Apple	519	415	52	52
Berry	340	272	34	34
Fig	508	406	50	52
Guava	520	416	52	52
Orange	547	437	55	55
Plum	468	374	47	47
Persimmon	527	421	53	53
Tomato	159	127	16	16
Subtotal	3,588	2,868	359	361
<i>Flowers (TFDS)</i>				
Daisy	633	506	64	63
Dandelion	898	718	90	90
Roses	641	512	64	65
Sunflowers	699	560	70	69
Tulips	799	639	80	80
Subtotal	3,670	2,935	368	367
<i>Flowers Recognition</i>				
Daisy	764	611	76	77
Dandelion	1052	842	105	105
Roses	784	627	78	79
Sunflowers	733	586	73	74
Tulips	984	787	98	99
Subtotal	4,317	3,453	430	434

5.5. Perturbation methods

To evaluate the robustness of the proposed framework under distribution shifts and common image corruptions, we have applied nine image perturbation methods to the test set. **if anything in the table is missing below, add it in the corresponding bullet** The perturbation parameters have been fixed across all experiments to ensure reproducibility and enable fair comparisons among different CNN architectures. The details of the perturbations are given below: **missing references here. Add 1 or more references to papers applying these for the same reason? i.e., evaluation under perturbation**

- **Gaussian Blur:** Applied using a 5×5 kernel to simulate the loss of fine spatial details caused by defocus or low-quality imaging conditions.
- **Gaussian Noise:** Additive Gaussian noise with zero mean and standard deviation 0.05 has been introduced to model sensor noise and random intensity fluctuations.
- **Brightness Adjustment:** Increased by adding a constant value of 0.2 to normalized pixel intensities, after which values have been clipped to the valid range $[0, 1]$.

- **Hue Shift:** Hue values in the HSV color space have been shifted by 15 units while preserving the saturation and value channels, simulating color variations caused by illumination or camera characteristics.
- **Contrast Adjustment:** Modified using mean-centered linear scaling: $I' = (I - \mu_I) \times 0.5 + \mu_I$, where I denotes the input image and μ_I represents the mean pixel intensity (computed per channel). The resulting values have been clipped to the range $[0, 1]$.
- **Gamma Correction:** Nonlinear illumination changes have been simulated using gamma correction: $I' = I^{2.0}$, where pixel intensities have been raised to the power of $\gamma = 2.0$.
- **Motion Blur:** Simulated using a 7×7 horizontal averaging kernel to emulate camera or object motion during image acquisition.
- **JPEG Compression:** Artifacts have been introduced by encoding and decoding the images with a quality factor of 30, simulating degradation due to lossy storage or transmission.
- **Random Occlusion:** A randomly positioned rectangular region corresponding to 20% of the image height and 20% of the image width has been masked by setting its pixel values to zero, thereby simulating partial object occlusion.

5.6. Training Configurations

We used TensorFlow/Keras on the Kaggle [cite kaggle](#) environment with two NVIDIA Tesla T4 GPUs to training and evaluation of all the CNN models in our experiments. The training configuration has followed the settings adopted in the original EgyPLI study[10]. **what are these? please specify. is it only for EgyPLI or all 4 datasets??** The custom CNN was trained from scratch, while other models have been initialized with ImageNet-pretrained weights [cite here](#) and their backbone layers have been frozen.

For the pretrained models, the original classification heads have been removed and replaced with a common classification head consisting of Global Average Pooling [cite](#), a dropout layer [cite](#) with dropout rate 0.3, a fully connected layer with 128 ReLU units [cite](#), and the final softmax layer has been configured according to the number of classes in each dataset. The Adam optimiser [cite](#) has been used with a learning rate of 5×10^{-5} and categorical cross-entropy loss [arg1](#) in all model optimizations.

6. Result and Discussion

6.1. Grad-CAM Visualization

6.1.1. Threshold Justification

6.2. Explainable Trustworthiness Index (ETI)

6.3. Explainable Robustness Index (ERI)

EgyPLI. XXXXXX

6.3.1. *BDMediLeaves*

6.3.2. *Flowers*

6.3.3. *Flowers Recognition KAGGLE Dataset*

6.4. Cross-Dataset Analysis

7. Related Works

Javed et al. introduced two approaches: a Bayesian approximation via Monte Carlo Dropout and the nonparametric Conformal Prediction framework to measure uncertainty [22]. Although CNN models are excellent for classification, their predictions are often wrong but overconfident. If wrong predictions receive high confidence, the models are overconfident and difficult to trust. These models suffer from poor calibration. Their research indicates that accuracy alone cannot make a model reliable. Recent studies have proposed trustworthiness metrics based on uncertainty estimation and failure detection. Ao et al. [23] proposed the Excess Area Under the Optimal Risk-Coverage Curve, E-AUoptRC, to evaluate model performance under the risk-coverage (RC) framework. They also introduced the Trust Index (TI), which displays model performance at the optimal coverage point and serves as a complementary metric to overall accuracy. E-AUoptRC and TI provide more trustworthiness of CNN models than other indices/metrics. Nastoska et al. proposed frameworks and metrics to evaluate AI trustworthiness, including fairness, transparency, privacy, and security [24]. They experimented with various frameworks such as the NIST AI Risk Management Framework (AI RMF), the AI Trust Framework and Maturity Model (AI-TMM), and ISO/IEC standards. Their study includes a comparative analysis of existing standards and their applications in the industry to describe their about generalizing effectiveness. Pal et al. proposed the Trust-Aware Explainable Artificial Intelligence (TAXAI) framework. It introduces a quantitative Trust Index for evaluating explainable AI systems in clinical applications [25]. The framework combines algorithmic transparency, interpretability alignment, and compliance-oriented reliability into a unique trust evaluation model. It is not similar to traditional XAI methods that primarily generate explanations. In contrast, TAXAI focuses on measuring the trustworthiness and reliability of explanations through a normalized trust score.

Zhang et al. proposed AR2, a repair method to improve robustness to corruption without changing the architecture. It aligns the class activation maps (CAMs) for clean and corrupted images and encourages the model to maintain consistency [26]. Although AR2 enhances robustness under perturbations, it does not provide any explainable score or index under various perturbations. In contrast, our proposed Explainable Robustness Index (ERI) directly measures the stability of Grad-CAM explanations under perturbations. It provides an explainability-based robustness evaluation. Hendrycks and Dietterich [27] introduced ImageNet-C and ImageNet-P. Their research evaluates performance under corruption and perturbations. They found negligible changes under corruption from AlexNet to ResNet classifiers. This

research calculates performance degradation under distribution shifts but does not assess the stability of model explanations. Su et al. evaluated the robustness of deep image classification models against various perturbations and demonstrated that high classification accuracy does not necessarily imply high robustness [28]. This work motivated us to investigate robustness beyond predictive accuracy. Taori et al. defined robustness during natural distribution shifts and emphasized that robustness should be evaluated separately from standard predictive accuracy [29]. This work further motivated us to evaluate robustness beyond predictive accuracy. Hendrycks et al. proposed AugMix, a data-processing method designed to improve the robustness of image classification models under common corruptions while maintaining performance on clean images [30]. This research is very motivating for our evaluation of robustness beyond standard predictive accuracy. Schneider et al. evaluated robustness under common image corruptions and showed that covariate-shift adaptation using corrupted test samples can substantially improve model performance under such distribution shifts [31]. Guo et al. proposed a model robustness evaluation framework [32]. It contains 23 comprehensive and rigorous metrics. These metrics consider two key aspects of adversarial learning: data and model.

Bassi et al. demonstrated the impact of background on the model's decision [11]. Background may cause shortcut learning. Layer-wise Relevance Propagation (LRP) explores the model's decision. They revealed that background bias influence on deep learning models can be minimized by LRP heatmaps without increasing computation cost. By injecting synthetic bias in the image's background, they compared their approach to eight state-of-the-art DNNs. Their approach remained more robust. Grad-CAM and saliency-based methods have been widely utilized to interpret CNN-based classification systems in various sectors, such as plant disease detection. They provide trustworthiness and important insights for decision-making and predictions [33]. Attention-based and explainability-guided CNN architectures have been widely explored in recent applications. For example, Singh et al. have proposed CBAM-enhanced VGG16 models for plant leaf disease classification. They presented the effectiveness of their research through comprehensive evaluation and interpretability analysis, such as Grad-CAM, Grad-CAM++, and Layer-wise Relevance Propagation (LRP), to make the system more transparent and reliable for smart farming. [34]. Shawon et al. highlight the growing application of explainable AI techniques such as Grad-CAM, SHAP, and LIME in plant disease detection. They emphasize concerns about generalizing models under real-world conditions and environmental variability, although they provide high accuracy [35]. However, these works primarily focus on applying interpretability methods rather than quantitatively evaluating explanation quality or robustness, which motivates the need for metrics such as ERI and ETI. A significant problem of Explainable AI is the unavailability of quantifiable metrics to evaluate or measure explanation quality. Rahimiaghdam and Alemdar

addressed the issue and proposed evaluative and quantitative metrics of explanation quality in medical imaging [36]. Baniecki. et al. expressed the quality of classifiers as the consistency of explanations and ground truth/segmentation masks. They proposed a framework that evaluates robustness by combining deep neural networks with explanations. Similar tasks are widely used in medical imaging. They introduced a fine-tuning approach that aligns model–explanation pipelines with ground truth. [37]. Repetto et al. evaluated the robustness of ExAI methods under various perturbations. They systematically measured the stability of ExAI techniques against both natural noise and adversarial manipulations that change explanations but not predictions. Finally, they introduced some evaluative metrics and proposed a protocol to assess the reliability of ExAI methods[38]. Zheng et al. proposed a spurious bias mitigation framework, ShortcutProbe[39]. Here, there is no need for annotation and labeling. ShortcutProbe detects prediction shortcuts that may lessen robustness. The models were retrained again to become invariant to these shortcuts and more robust. Guidotti et al. propose an approach that allows evaluating the local explanation methods using ground-truth explanations [40]. We have been motivated by this ground-truth-based evaluation perspective to quantify the spatial alignment between Grad-CAM evidence and ground-truth object regions, leading to the proposed Localization Mask Overlap Score (LMOS). Fong and Vedaldi proposed a general framework that can learn different kinds of explanations for any black-box algorithm and another framework to find the image region most responsible for a classifier decision [41]. Kim et al. attempted to find out the gap between classification and localization in weakly supervised object localization and showed that CAM-based methods tend to emphasize the most discriminative object parts rather than the complete object region [42].

8. Conclusion

Data and Code availability

Supplementary code and data (including .json files and .keras files) associated with this research are available at:

https://github.com/ragib1996/ETI_ERI

CRedit authorship contribution statement

Ragib Mehedi: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing – Original Draft, Writing – Review & Editing. **Azmain Inquaid Haque:** Investigation, Data Curation, Software, Validation. **Md. Nazmul Abdal:** Data Curation. **Abdullah Al Noman:** Supervision, Writing – Review & Editing. **Md. Zahidul Islam:** Supervision, Writing – Review & Editing .

Acknowledgments

Disclosure statement

Funding

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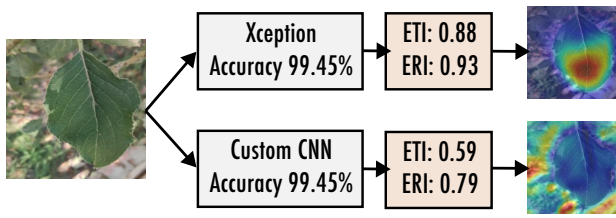


Figure 1: Comparison between traditional accuracy and the proposed XAI-based metrics (ETI and ERI). Although both Xception and the custom CNN achieve identical 99.45% validation accuracy, Grad-CAM heatmaps reveal that the custom CNN relies on background context. The proposed ETI and ERI metrics successfully capture this flaw by producing lower scores for the custom model.